Kenyan Multilingual PSA Machine Translation

Translating Public Service Announcements between English/Kiswahili and under-resourced Kenyan languages — Ekegusii, Somali, and Dholuo.

DSA 4020A — Natural Language Processing · Semester Project · Supervised by Dr. Edward Ombui
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4 languages | 29,609 rows | 5 domains

INTRODUCTION

A Public Service Announcement is a short, action-oriented message issued by a government agency, NGO, or media outlet to inform, warn, or guide the public: a Ministry of Health advisory about a disease outbreak, an electoral commission's reminder to verify voter registration, a county government's drought warning. When these messages are only available in English or Kiswahili, speakers of Kenya's other language communities are functionally excluded from information meant to reach everyone.

This project's objective was to build and evaluate a proof-of-concept translation system covering four target languages across five PSA domains, and deploy a working demonstration, not merely a research result on paper. Ekegusii, Somali, and Dholuo were chosen because they represent three different points on the resource-availability spectrum, which turned out to matter for how the project unfolded.

Two modeling strategies were run in parallel and reported in full, on a shared metric wherever the raw outputs allowed recomputing it, rather than committing early to one and hiding the other: pretrained-model fine-tuning (mT5/NLLB) and dictionary-prompted LLM translation with zero fine-tuning.

PRIOR WORK & THE RESOURCE GAP

Somali and Dholuo both have real coverage in NLLB-200, Meta's 200-language open translation model, and in the FLORES-200 evaluation benchmark that accompanies it. Ekegusii has neither: it is not one of NLLB's 200 supported languages, and no major open translation model has ever been trained on it. This single fact, confirmed early and treated as a hard constraint rather than an inconvenience, shaped the entire second half of the project.

Broader multilingual and African-NLP efforts were checked against this specific gap before any modeling began. MAFAND-MT and the Masakhane initiative advance machine translation for African languages generally and were consulted as the closest methodological precedent, but neither targets this project's specific combination: five PSA domains, three under-resourced Kenyan languages, and Ekegusii in particular. No existing system was found that addresses this combination directly.

METHODOLOGY

1. Pretrained fine-tuning. mT5-small and NLLB-200-distilled-600M, few-shot fine-tuned for English-to-Kiswahili and English-to-Ekegusii. Adafactor optimiser, lr 1e-3, 3 epochs, batch 16, 4,000 examples/direction cap in the original light run (a full-scale rerun, 5 epochs on the complete dataset, was added this pass -- see notebooks/full_pipeline/). Splits assigned by PSA identifier so no announcement's sentences leak across train/dev/test. Tracked with MLflow throughout. Because NLLB-200 has no native Ekegusii code, the standard workaround is to repurpose a linguistically related language's tag -- a legitimate technique with precedent, but exactly the kind of decision that can quietly bias a result if left unstated, so it is recorded here.

2. Dictionary-prompted LLM, zero fine-tuning. at inference time the system retrieves the three most relevant real example translations from a 13,497-row bank (agriculture PSAs, Bible text for everyday conversational vocabulary, a five-domain corpus added to fix a coverage gap) via TF-IDF nearest-neighbour search. A three-tier bilingual dictionary lookup follows: exact match against a 95,418-word dictionary, falling back to a morphological lemma match, falling back again to a WordNet-synonym match at explicitly lower confidence. Kenyan county names and institution acronyms are protected by a dedicated entity-recognition pass. Degenerate output triggers up to four retries with a reinforced prompt and escalating temperature before an honest flag. This mechanism was arrived at after fine-tuning was tried first and found wanting, not chosen from the outset.

RESULTS

Somali 83.1 chrF: NLLB-200 zero-shot, clearly ahead of a cloud translation API's 65.8 on a 40-row real gold sample.

Kiswahili 80.7: cloud translation API in production, NLLB-200 (67.7) as automatic fallback for the ~31% of a 15,000-row run the API's quota couldn't cover.

Ekegusii 54.8: dictionary-prompted, 70B-class backend -- required an entirely different technique from the two languages above, since NLLB has never seen Ekegusii.

Dholuo 53.7: NLLB-200 zero-shot, unambiguous winner over every LLM backend tested (16-23 chrF), several of which produced output not recognisable as Dholuo at all.

Every method scored on the identical 52 held-out real agriculture Ekegusii sentences, identical scoring code -- the one genuinely fair head-to-head in the project.

Zero-shot baseline: 7.1 BLEU, ~15-21 chrF, 0.01-0.02 recall.

mT5, fine-tuned (light run): 5.8 BLEU, 26.2 chrF.

Fine-tuned NLLB, best: 17.8 chrF, 0.81 recall.

Dict-prompted + morphology: 29.4 BLEU, 51.1 chrF, 0.92 recall.

Dict-prompted, 70B backend: 34.0 BLEU, 54.8 chrF, 0.90 recall, best result on this test.

GENERALIZATION AND PRODUCTION REPAIR

Cross-domain fix: validated on agriculture alone the mechanism reached 45-55 chrF; tested on real Education/Health/Security/Governance sentences it dropped to 26.6. Widening the retrieval bank from agriculture+Bible to a full five-domain corpus (13,497 rows) raised it to 38.6 chrF, a real +12-point gain, narrowed not closed: still below the agriculture range.

Production regression: a compute-constrained simplified prompt path caused 3.9% degenerate output on a 10,169-row batch. Root-caused to missing few-shot grounding, not just re-run with more compute. A fresh scan found 131 further candidates (9 genuine failures, 78 fragmented-source rows, 44 false positives); the 482-row repair batch came back 96.9% (467/482) clean across four escalating-temperature retry rounds, the remaining 15 honestly flagged rather than force-accepted. Full forensic detail: METHODOLOGY & REGRESSION, IN FULL DETAIL, below.

INTERPRETATION AND QUALITY ASSURANCE

Why prompting won out: found independently twice, by two different people, on two different languages. On Ekegusii, best fine-tuned chrF 17.8 vs. best prompted 54.8. Separately, a teammate's own Kiswahili fine-tuning attempt found BLEU drop from 75.4 zero-shot to 3.9 after fine-tuning, a catastrophic-forgetting pattern consistent with Ekegusii. That said, this used a light fine-tuning config -- at full scale, fine-tuned NLLB reaches 40.3 chrF, close to prompting's own cross-domain 38.6, narrowing the gap considerably rather than losing outright (see COMET & FINE-TUNING AT SCALE below).

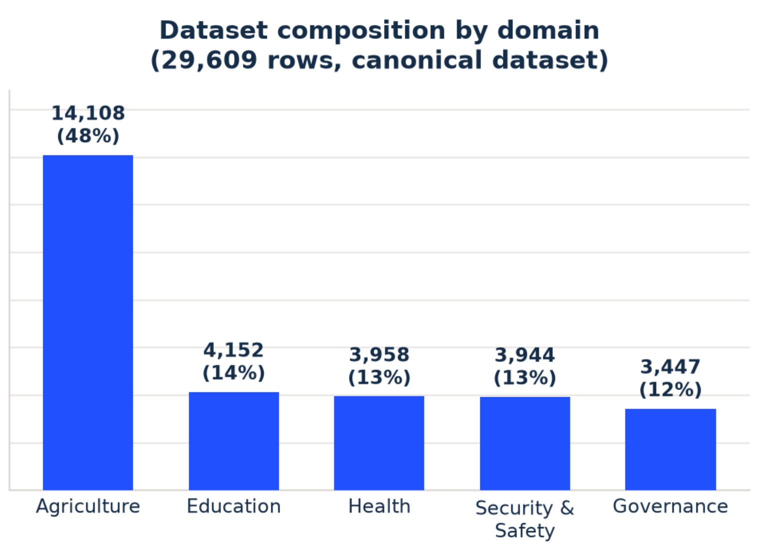
Manual QA caught what automated metrics missed: round-trip back-translation scoring, used to pick the best of several candidate translations, was checked against manual reads and found to twice select the objectively worst candidate -- once, a correct translation of a technical term scored lowest because back-translation drifted less for a wrong answer than the right one. Separately, a teammate's Kiswahili submission had a systematic error (one institution's name garbled in 181 of 200 rows) that an automated score alone had not surfaced; it was retranslated from scratch rather than silently kept.

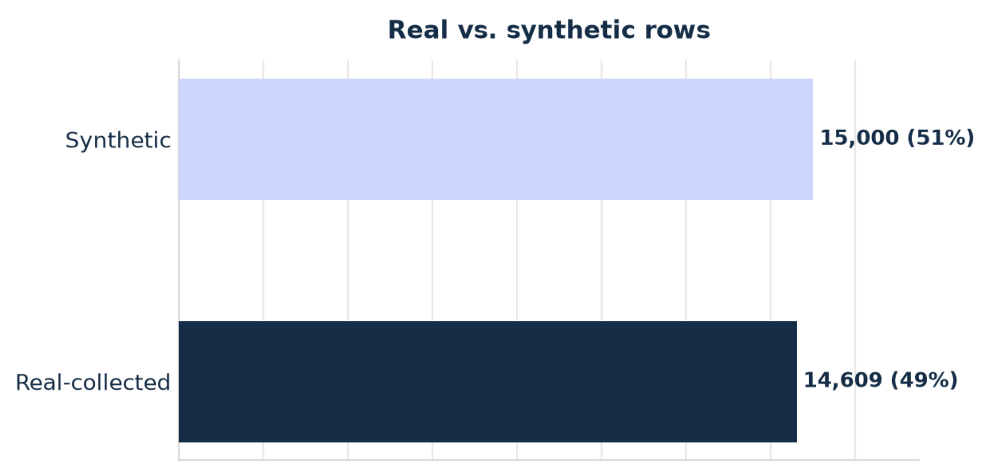
LIMITATIONS & FUTURE WORK

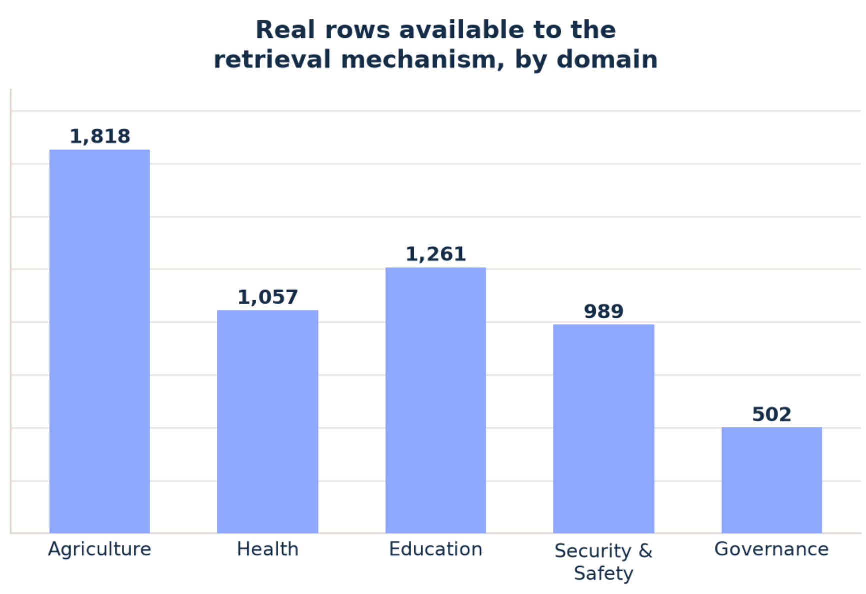
Limitations, stated honestly
- Scored native-speaker evaluation not yet run at scale (both approaches)
- Cross-domain gap narrowed and root-caused per domain, not closed
- A shared university GPU platform deployment was attempted, blocked on an unresolved vLLM engine-core crash
- Degeneracy detection can still be tightened

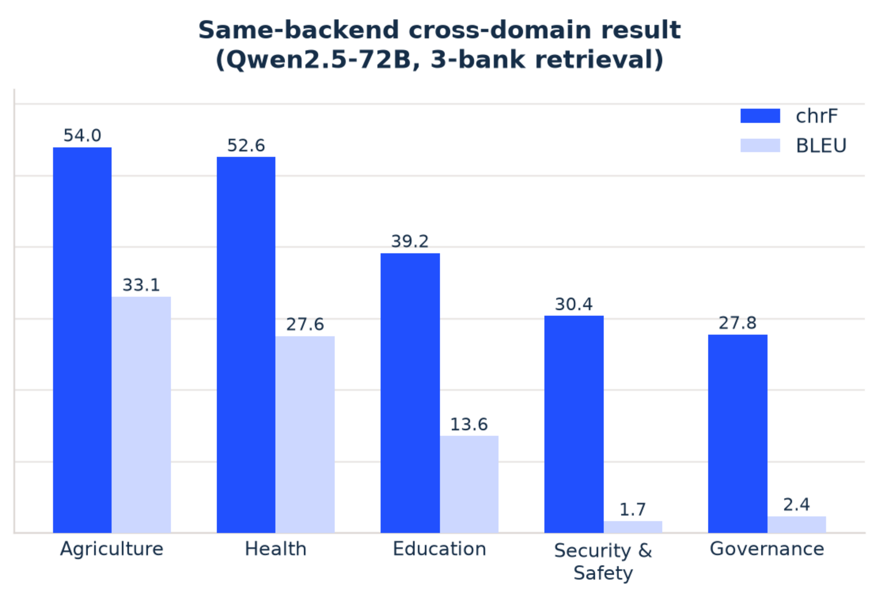
Future work, already scoped
- Run a scored native-speaker evaluation pass (100+ sentences)
- Governance-only synthetic-data augmentation ablation (the one domain the data actually supports)
- Apples-to-apples fine-tuning-vs-prompting re-test, same eval set both sides
- Resolve the shared-GPU-platform vLLM crash and complete the public-facing deployment

DATASET COMPOSITION & DOMAIN PERFORMANCE







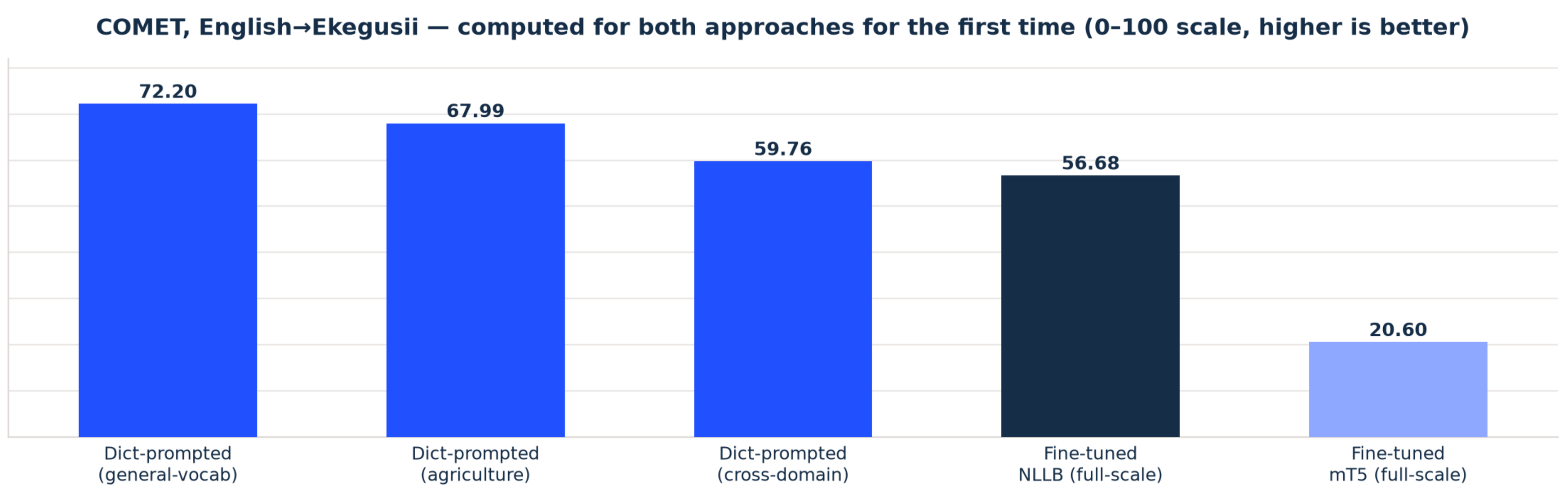


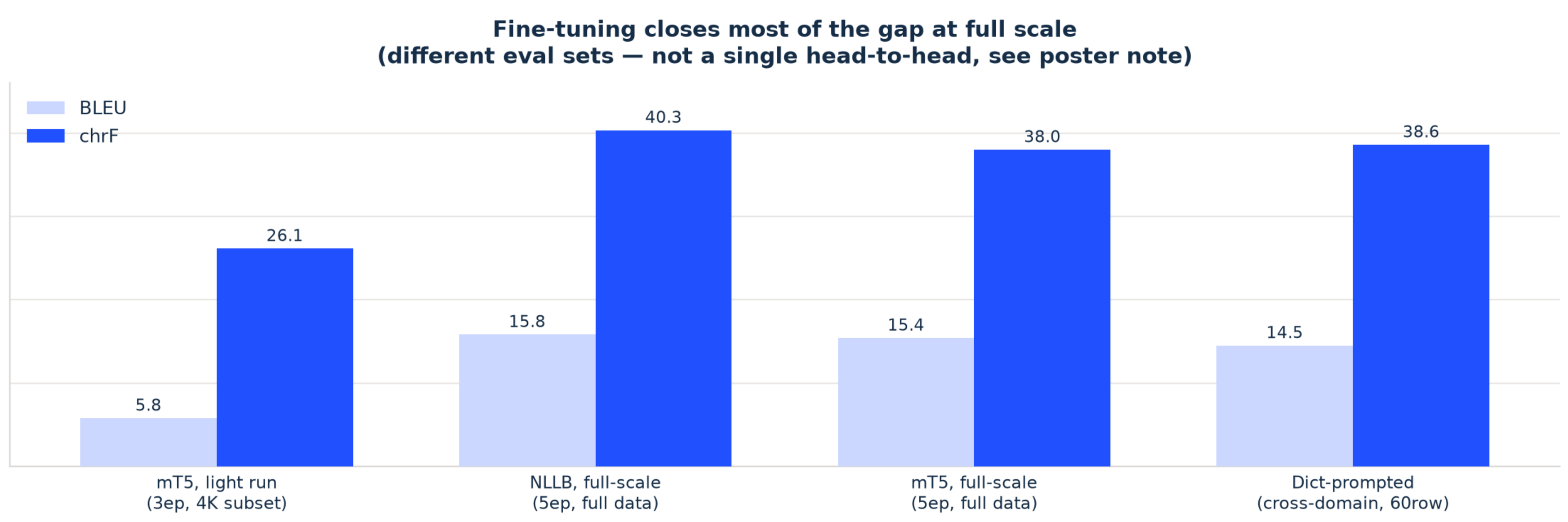
Governance: least data by far (502 rows vs. 989-1,261 elsewhere) -- a genuine thin-data case, real candidate for synthetic augmentation.

Education: more data than Health/Security yet scores worse -- lowest recall (0.75) points to a dictionary-coverage gap, not a volume problem.

Security & Safety: perfect recall (1.00) but worst BLEU (1.7) -- right words, poor fluency/structure. Not a data-availability problem at all. Agriculture is 48% of the whole dataset, not just the retrieval bank -- the single largest driver of its lead across every metric.

COMET & FINE-TUNING AT SCALE





COMET computed for both approaches for the first time -- dictionary-prompting leads (56.68-72.20) over fine-tuning (20.60-56.68), though gaps aren't uniform. The original mT5 fine-tuning comparison (5.8 BLEU / 26.2 chrF) used a light 3-epoch, 4,000-row run; trained at full scale instead (5 epochs, full dataset, notebooks/full_pipeline/), fine-tuned NLLB reaches 40.3 chrF -- close to prompting's own cross-domain result. Prompting still leads on the agriculture-focused comparison this project ran most, but at scale, fine-tuning is competitive, not simply behind.

METHODOLOGY & REGRESSION, IN FULL DETAIL

Retrieval and dictionary: TF-IDF nearest-neighbour search over a 13,497-row bank (agriculture PSAs, Bible text, five-domain corpus) retrieves three real examples per sentence; a three-tier dictionary lookup (exact match, morphological lemma, WordNet-synonym fallback at lower confidence) supplies content-word constraints.

Protection and retry: county names and institution acronyms are protected by a dedicated entity-recognition pass; degenerate output triggers up to four retries with a reinforced prompt and escalating decoding temperature before an honest flag rather than a silent bad translation.

The 131-candidate breakdown: a fresh full-dataset scan beyond the original 395 flagged rows found 131 more candidates, splitting on manual review into 9 genuine additional repetition failures, 78 rows where the English source itself was a fragmented excerpt (a different r
false positives confirmed fluent and left untouched.

Repair result: the 482-row repair batch, re-run through the complete mechanism across four escalating-temperature retry rounds, came back 467/482 (96.9%) clean; the remaining 15 were left honestly flagged rather than force-accepted.